

Generation Interconnection Request System Impact Study Report

For Queue Project Q15-0509

Novec Paradise 138 kV
June 25, 2016

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q15-0509. The Interconnection Customer proposes a 262.1 MW battery energy storage facility interconnecting at the Novec Paradise 138 kV substation in the Virginia area. The request was received on August 16, 2015. The study evaluated system intact and single contingency conditions using the regional planning case, including 301 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 36 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$91,014,000.

3. Network Impacts

3.1 Generator Deliverability and Single Contingency Analysis

Monitored Facility	Contingency	Loading (%)	Rating (MVA)	DFAX (%)
GRAND BLANC GENERATING STATION to WALNEY 138kV	GRAND BLANC GENERATING STATION to DEL AMO 138kV	183.8	210	12.6
NOVEC PARADISE to PASCO CNTY SOLID WASTE RESOURCE RECOVERY 138kV	CHAFFEE GAS RECOVERY to S JORDAN 138kV line	175.9	240	42.7
GRAND BLANC GENERATING STA to CUB-RUN DP 138kV	LUNA to YAVAPAI 138kV	167.4	346	15.2
SULLY to WALNEY 138kV line	DEL AMO to GRAND BLANC GENERATING STA 138kV	155.1	226	12.6
THOMSTON to TOA VACA 138kV	BLAINE to CHAFFEE GAS RECOVERY 138kV	154.5	159	25.1
S JORDAN to CHAFFEE GAS RECOVERY 138kV	WHEELER to FIREBANTZ 138kV	151.9	263	9.7
NOVEC PARADISE to DEAN DRIVE AREA ELECTRIC GENERATORS SUBSTATION 138kV line	BLAINE to CHAFFEE GAS RECOVERY 138kV Line	148.9	244	44.5
DEAN DRIVE AREA ELECTRIC GENERATORS to W STAUNTON 138kV	CHAFFEE GAS RECOVERY to BLAINE 138kV	148.9	244	44.5
WOODS DP to WEST STAUNTON 138kV Line	CHAFFEE GAS RECOVERY to BLAINE 138kV	143.9	324	33.3
TOA VACA to WOODS DP 138kV Line 2	WOODS DP to TOA VACA 138kV #1	142.6	331	26.4
BULL RUN to DEL AMO 138kV	GRAND BLANC GENERATING STA to WALNEY 138kV Line	140.5	226	8.0

Monitored Facility	Contingency	Loading (%)	Rating (MVA)	DFAX (%)
FOUNTAIN to THOMASTON 138kV	GRAND BLANC GENERATING STA to CUB-RUN DP 138kV	139.2	316	11.8
FILLMORE to BULL RUN 138kV	WALNEY to GRAND BLANC GENERATING STATION 138kV	135.2	219	8.0
CASTLETON ENERGY CTR to SULLY 138kV	DEL AMO to GRAND BLANC GENERATING STA 138kV	131.6	249	12.6
WOODS DP to VMEA PEAKING GEN 138kV	WOODS DP to W STAUNTON 138kV	130.6	234	12.6
BLAINE to STONEWALL ENERGY CTR 138kV L	NOVEC PARADISE to DEAN DRIVE AREA ELECTRIC GENERATORS 138kV	130.1	230	25.3
FIREBANTZ to WHEELER 138kV	S JORDAN to CHAFFEE GAS RECOVERY 138kV	129.4	349	10.3
MOUNT PLEASANT to MIDDLETOWN JUNCTION 138kV	LEWISBURG to S JORDAN 138kV line	129.2	220	5.6
LUNA to YAVAPAI 138kV line	GRAND BLANC GENERATING STATION to CUB-RUN DP 138kV	129.0	246	9.3
LEWSBURG to S JORDAN 138kV	REMINGTON CT to RIVERTON 230 138kV	128.7	285	6.5
TOA VACA to WOODS DP 138kV	WOODS DP to TOA VACA 138kV L	126.8	372	26.4
RIVERTON 230 to REMINGTON CT 138kV line	LOUISA GENERATION FACILITY to CARRABELLE 345kV	126.5	499	5.3
BLAINE to CHAFFEE GAS RECOVERY 138kV line	NOVEC PARADISE to DEAN DRIVE AREA ELECTRIC GENERATORS 138kV	125.8	264	25.3
PEAVY FALL HY to LUNA 138kV line	GRAND BLANC GENERATING STA to CUB-RUN DP 138kV	123.8	228	9.3
CLOVER HILL to PRINCE WILLIAM DP 138kV	WEST STAUNTON SUBSTATION to WOOS DP 138kV	123.4	232	13.0
CLOVER HILL to ELK HORN 138kV Line	WOODS DP to WEST STAUNTON 138kV line	123.4	232	13.0
FAIRLAND to SIUC 138kV	TABER to GLENANNAN 345kV	122.5	832	5.2
FLUVANNA P.S to LOUISA GENERATION FACILITY 345kV	LEAVITT SUBSTATION to BIRCHWOOD 345kV	120.3	2,485	6.0
DEL AMO to GRAND BLANC GENERATING STATION 138kV	GRAND BLANC GENERATING STA to WALNEY 138kV	120.3	300	8.0
STONEWALL ENERGY CENTER to FOUNTAIN 138kV Line	NOVEC PARADISE to DEAN DRIVE AREA ELECTRIC GENERATORS 138kV line	120.1	223	25.3
MOUNT PLEASANT to FIREBANTZ 138kV Line	CHAFFEE GAS RECOVERY to S JORDAN 138kV	119.6	289	7.6
ELK HORN to HOOVER 138kV	W STAUNTON SUBSTATION to WOODS DP 138kV	118.7	227	13.0
HOOVER to MENAN 138kV	WEST STAUNTON SUBSTATION to WOODS DP 138kV	117.9	229	13.0
LEWISBURG to SILOAM SPRINGS 138kV Line	INDECK W ENFIELD ENERGY CTR to ACADIANA MALL 138kV	113.7	246	5.9
CHAFFEE GAS RECOVERY to MICHIGAN WIND 2 138kV	BLAINE to CHAFFEE GAS RECOVERY 138kV line	111.0	193	11.3

Monitored Facility	Contingency	Loading (%)	Rating (MVA)	DFAX (%)
W STAUNTON to PRINC WILLIAM DP 138kV	PASCO CNTY SOLID WASTE RESOURCE RECOVERY to NOVEC PARADISE 138kV	108.8	298	28.0

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
GRAND BLANC GENERATING STATION to WALNEY 138kV L	Rebuild line with higher-capacity conductor	\$3,025,000	100% (\$3,025,000)
NOVEC PARADISE to PASCO CNTY SOLID WASTE RESOURCE RECOVERY 138kV	Rebuild line with higher-capacity conductor	\$2,186,000	68% (\$1,475,000)
CUB-RUN DP to GRAND BLANC GENERATING STATION 138kV	Rebuild line with higher-capacity conductor	\$2,428,000	100% (\$2,428,000)
SULLY to WALNEY 138kV line	Rebuild line with higher-capacity conductor	\$2,021,000	100% (\$2,021,000)
THOMASTON to TOA VACA 138kV line	Rebuild line with higher-capacity conductor	\$2,582,000	43% (\$1,099,000)
S JORDAN to CHAFFEE GAS RECOVERY 138kV	Rebuild line with higher-capacity conductor	\$18,547,000	100% (\$18,547,000)
NOVEC PARADISE SUBSTATION to DEAN DRIVE AREA ELECTRIC GENERATORS 138kV Line	Rebuild line with higher-capacity conductor	\$4,198,000	72% (\$3,014,000)
DEAN DRIVE AREA ELECTRIC GENERATORS to W STAUNTON 138kV	Rebuild line with higher-capacity conductor	\$841,000	72% (\$604,000)
WEST STAUNTON SUBSTATION to WOODS DP SUBSTATION 138kV	Rebuild line with higher-capacity conductor	\$781,000	48% (\$376,000)
TOA VACA to WOODS DP 138kV Line 2	Rebuild line with higher-capacity conductor	\$1,611,000	42% (\$677,000)
BULL RUN to DEL AMO 138kV	Rebuild line with higher-capacity conductor	\$2,775,000	100% (\$2,775,000)
FOUNTAIN to THOMASTON 138kV	Rebuild line with higher-capacity conductor	\$978,000	100% (\$978,000)
FILLMORE to BULL RUN 138kV	Rebuild line with higher-capacity conductor	\$5,770,000	100% (\$5,770,000)
SULLY to CASTLETON ENERGY CENTER 138kV Line	Rebuild line with higher-capacity conductor	\$2,373,000	100% (\$2,373,000)
WOODS DP to VMEA PEAKING GEN 138kV	Rebuild line with higher-capacity conductor	\$4,803,000	100% (\$4,803,000)
BLAINE to STONEWALL ENERGY CTR 138kV L	Rebuild line with higher-capacity conductor	\$1,704,000	100% (\$1,704,000)

Facility	Required Upgrade	Cost Estimate	Allocation to Project
WHEELER SUBSTATION to FIREBANTZ 138kV	Rebuild line with higher-capacity conductor	\$5,479,000	13% (\$703,000)
MIDDLETOWN JCT to MOUNT PLEASANT 138kV	Rebuild line with higher-capacity conductor	\$7,023,000	100% (\$7,023,000)
LUNA to YAVAPAI 138kV line	Rebuild line with higher-capacity conductor	\$1,481,000	100% (\$1,481,000)
LEWSBURG to S JORDAN 138kV	Rebuild line with higher-capacity conductor	\$8,895,000	22% (\$1,968,000)
TOA VACA to WOODS DP 138kV	Rebuild line with higher-capacity conductor	\$1,090,000	42% (\$458,000)
RIVERTON 230 to REMINGTON CT 138kV	Rebuild line with higher-capacity conductor	\$27,579,000	5% (\$1,460,000)
BLAINE to CHAFFEE GAS RECOVERY 138kV	Rebuild line with higher-capacity conductor	\$1,611,000	100% (\$1,611,000)
PEAVY FALL HY to LUNA 138kV line	Rebuild line with higher-capacity conductor	\$2,081,000	100% (\$2,081,000)
CLOVR HILL to PRINCE WILLIAM DP 138kV	Rebuild line with higher-capacity conductor	\$3,113,000	100% (\$3,113,000)
ELK HORN to CLOVER HILL 138kV	Rebuild line with higher-capacity conductor	\$1,434,000	100% (\$1,434,000)
SIUC to FAIRLAND 138kV Line	Rebuild line with higher-capacity conductor	\$1,310,000	4% (\$51,000)
FLUVANNA P.S to LOUISA GENERATION FACILITY 345kV	Rebuild line with higher-capacity conductor	\$72,067,000	2% (\$1,047,000)
DEL AMO to GRAND BLANC GENERATING STATION 138kV	Rebuild line with higher-capacity conductor	\$1,819,000	100% (\$1,819,000)
STONEWALL ENERGY CENTER to FOUNTAIN 138kV Line	Rebuild line with higher-capacity conductor	\$1,397,000	100% (\$1,397,000)
MOUNT PLEASANT to FIREBANTZ 138kV Line	Rebuild line with higher-capacity conductor	\$3,899,000	32% (\$1,234,000)
HOOVER to ELK HRN 138kV line	Rebuild line with higher-capacity conductor	\$927,000	100% (\$927,000)
HOOVER to MENAN 138kV	Rebuild line with higher-capacity conductor	\$1,254,000	100% (\$1,254,000)
LEWISBURG to SILOAM SPRINGS 138kV Line	Rebuild line with higher-capacity conductor	\$8,648,000	100% (\$8,648,000)
CHAFFEE GAS RECOVERY to MICHIGAN WIND 2 138kV line	Rebuild line with higher-capacity conductor	\$1,285,000	100% (\$1,285,000)
W STAUNTON to PRINC WILLIAM DP 138kV	Reconductor line	\$351,000	100% (\$351,000)
	Total allocated to project		\$91,014,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.